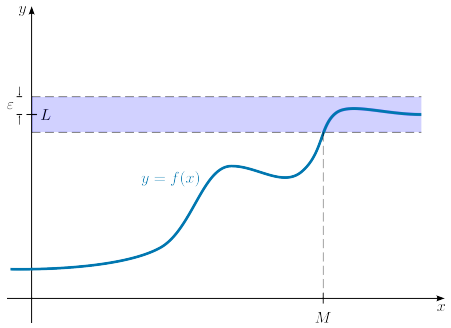
Definition 0.1 (Limit as $x \to \infty$). Let f be defined on $[c, \infty)$ for some $c \in \mathbb{R}$ We say that $\lim_{x\to\infty} f(x) = L$ if for each $\varepsilon > 0$ there exists a corresponding $M > 0$ such that for all $x > M$, we have

$$|f(x) - L| < \varepsilon$$

Figure 1: Visual representation of the definition of $\lim_{x\to\infty} f(x) = L$



Example. Consider the function $f(x) = \dfrac{1}{x}$ for $x > 0$. We want to show that $\lim_{x\to\infty} f(x) = 0$.

Proof: Let $\varepsilon > 0$ and choose $M = \dfrac{1}{\sqrt{\varepsilon}}$. Then, if $x > M$, we have

$$\begin{aligned}
|f(x) - 0| &= \left|\frac{\sin 3x}{x^2 + 4}\right| \\
&= \frac{1}{|x^2 + 4|} && \text{Since } |\sin 3x| \leq 1 \\
&\leq \frac{1}{x^2} && \text{Since } x^2 < x^2 + 4 \text{ as seen on this } \textit{graph} \\
&\leq \frac{1}{M^2} && \text{Since } x > M \\
&= \frac{1}{\left(\frac{1}{\sqrt{\varepsilon}}\right)^2} && \text{Substituting } M = \frac{1}{\sqrt{\varepsilon}} \\
&= \varepsilon && \square
\end{aligned}$$

Worked Example 0.1 Prove that $\lim_{x\to\infty} \dfrac{x}{1 + x^2} = 0$. **Proof:** Let $\varepsilon > 0$ and choose $M = \dfrac{1}{\varepsilon}$.

Then, if $x > M$, we have

$$\begin{aligned}
\left|\frac{x}{1+x^2} - 0\right| &= \frac{x}{1+x^2} \\
&\leq \frac{x}{x^2} && \text{Since } 1 + x^2 > x^2 \\
&= \frac{1}{x} \\
&< \frac{1}{M} && \text{Since } x > M \\
&= \varepsilon && \square
\end{aligned}$$

Exercise 0.1 Prove that $\lim_{x\to\infty} \dfrac{x}{x-3} = 1$.

Proof: Let $\varepsilon > 0$ and choose $M = \dfrac{3}{\varepsilon}$. Then, if $x > M$, we have

$$\begin{aligned}
\left|\frac{x}{x-3} - 1\right| &= \left|\frac{x-(x-3)}{x-3}\right| \\
&= \left|\frac{3}{x-3}\right| \\
&< \left|\frac{3}{M}\right| && \text{Since } x > M \\
&= \varepsilon && \square
\end{aligned}$$

Exercise 0.2 Prove that $\lim_{x\to\infty} \dfrac{2x-3}{x+2} = 2$.

$$\begin{aligned}
\left|\frac{2x-3}{x+2} - 2\right| &= \left|\frac{2x-3-2(x+2)}{x+2}\right| \\
&= \left|\frac{-7}{x+2}\right| \\
&< \left|\frac{7}{M}\right| && \text{Since } x > M \\
&= 7\varepsilon && \text{Since } M = \frac{7}{\varepsilon} \\
&= \varepsilon && \square
\end{aligned}$$

Exercise 0.3 Prove that $\lim_{x\to\infty} \dfrac{6x^2+5x}{2x^2-7} = 3$.

Proof: Let $\varepsilon > 0$ and choose $M = \dfrac{1}{\varepsilon}$. Then, if $x > M$, we have

$$\begin{aligned}
\left|\frac{6x^2+5x}{2x^2-7} - 3\right| &= \left|\frac{6x^2+5x-3(2x^2-7)}{2x^2-7}\right| \\
&= \left|\frac{6x^2+5x-6x^2+21}{2x^2-7}\right| \\
&= \left|\frac{5x+21}{2x^2-7}\right| \\
&< \left|\frac{5M+21}{2M^2-7}\right| && \text{Since } x > M \\
&= \frac{5M+21}{2M^2-7} && \text{Substituting } M = \frac{1}{\varepsilon} \\
&= \varepsilon && \square
\end{aligned}$$